Rudra Patel

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EDUCATION

University of Pennsylvania - Philadelphia, PA

BSE in Mechanical Engineering and Applied Mechanics *May 2030*

- **Relevant Classes:** Multivar. Calc., Physics 1: Eng. Mechanics, Physics 2: Elec. & Magnetism, Intro to Mech. Design, Gen. Chem 1
- **Relevant Extracurriculars:** Wharton Undergraduate Aerospace Club (Consulting Committee), American Society of Mechanical Engineers (ASME)

PROFESSIONAL EXPERIENCE

Penn Aerial Robotics - Philadelphia, PA *Sep. 2026 - Present*

Mechanical Team

- Co-designed and sized a monoplane from the ground up in XFLR5 for Micro Class; optimized for static and dynamic stability
- Sized wing chord, tail, and elevator from first principles; ran 5+ design iterations against sizing and budget requirements
- Root-caused instability by isolating CG and elevator variables; improved static stability by 2.6x while balancing drag and materials
- Simulated battery and motor thrust to optimize takeoff weight while accounting for battery sag and motor inefficiency

Penn Jet Propulsion - Philadelphia, PA *Sep. 2026 - Present*

R&D Team

- Designed nozzle in SolidWorks; traded 4 candidate materials against cost, strength, and manufacturability requirements
- Validated structural integrity of nozzle via SolidWorks FEA, achieving a high factor of safety; carried the design from requirements definition through qualification, finishing top 3 in the internal competition; delivered at 33% of allotted budget
- Analyzed competition nozzle and a test nozzle in Ansys Fluent (CFD) to compare pressure, temperature, and outlet velocity
- Researching flap vs. gimbal thrust vectoring, comparing advantages and disadvantages of each to help guide team direction
- Designing flap thrust vectoring mechanism in SolidWorks for a microturbojet; modeled flameholder and injector parts as training

Charge Robotics - Marlton, NJ *Sep. 2022 - June 2026*

Executive Board, Robot Operator, GitHub Project Manager

- Directed 5+ members across mechanical, electrical, and software subteams to integrate 5+ subsystems against system requirements
- Created 3 PID velocity and positional controllers with Java and tuned P, D, and FF variables to smoothly control all subsystems
- Increased performance of 4+ end effectors by maximizing motor and sensor use; onboarded 10+ new software members
- Diagnosed and fixed hardware failures between matches; spearheaded strategy sessions; led team to district finals as operator

Residential Light Usage Tracker - Marlton, NJ *Sep. 2025 - June 2026*

Capstone Project

- Designed light-switch-mounted actuation mechanism in Onshape (5+ CAD parts), taking it from requirements to prototype
- Programmed Arduino Nano-driven micro servo; durability-tested 6 iterations, root-causing each failure before final design
- Developed door sensor and companion app with a team of four to collect light usage data to reduce household energy costs

New Jersey Cybersecurity and Communications Integration Cell - Hamilton, NJ *July 2025 - Sep. 2025*

Cybersecurity Intern

- Researched security of aircraft systems, ground systems, and satellites; presented findings to agency leadership at intern conference
- Recommended policy changes to director and officials at HQ; co-led introduction of a cybersecurity newsletter with other interns

LEADERSHIP EXPERIENCE

India Temple Association - Berlin, NJ *Aug. 2022 - Aug. 2026*

Student Leader (Volunteer Teacher)

- Taught 55+ 6th graders Indian History/Values by creating fun/engaging educational content; volunteered 375+ hours
- Helped raise $5,000+ for various charities through annual class charity acts; mentored 2 younger high school student leaders

WORK EXPERIENCE

Rally House - Mount Laurel, NJ *June 2026 - Aug. 2026*

Sales Associate

- Delivered excellent customer service at the point of sale at the register; customer email capture rate was 85%+
- Created meaningful connections with customers when they entered and left the store; maintained knowledge of products

SKILLS AND INTERESTS

Programming: Python, Java, Visual Basic

Tools and Platforms: First Principles Engineering Design, SolidWorks (FEA, part/assembly), Ansys Fluent (CFD), Onshape, 3D Printing, Laser Cutting, Arduino, GitHub, VS Code, Claude Code, Microsoft Suite

Languages: English, Gujarati, Italian

Interests: Skiing, Aviation, Biking, Robotics, Cooking, Football